

Program No: 1

Date: 29/06/2026

Program Title: Create an app that displays name and email id.

MainActivity.java:

```
package com.example.labcycle;
import android.annotation.SuppressLint;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;
import android.widget.EditText;
import android.widget.Button;
import android.widget.TextView;

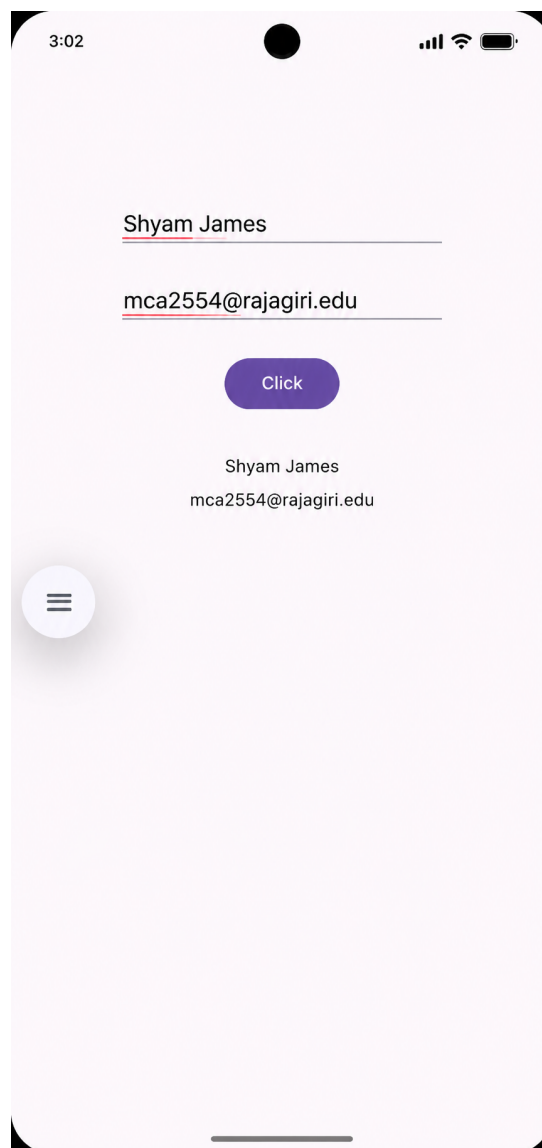
public class MainActivity extends AppCompatActivity {
    EditText name,email;
    Button btn;
    TextView t1,t2;
    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        name = findViewById(R.id.nametxt);
        email = findViewById(R.id.emailTxt);
        btn = findViewById(R.id.btnclick);
        t1 = findViewById(R.id.txtView1);
        t2 = findViewById(R.id.txtView2);
        btn.setOnClickListener(new View.OnClickListener(){
            @Override
            public void onClick(View view) {
                String n1 = name.getText().toString();
                String n2 = email.getText().toString();
                t1.setText(n1);
                t2.setText(n2);
            }
        });
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/nametxt"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:hint="Name"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="150dp"/>
    <EditText
        android:id="@+id/emailTxt"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:hint="Email"
        app:layout_constraintTop_toBottomOf="@id/nametxt"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp"/>
    <Button
        android:id="@+id/btnclick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click"
        app:layout_constraintTop_toBottomOf="@id/emailTxt"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="20dp"/>
    <TextView
        android:id="@+id/txtView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name"
        app:layout_constraintTop_toBottomOf="@id/btnclick"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="30dp"/>
```

```
<TextView
    android:id="@+id/txtView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Email"
    app:layout_constraintTop_toBottomOf="@id/txtView1"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="10dp"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 2

Date: 29/06/2026

Program Title: Create an app that takes marks of any three subjects from the user and displays its sum and average.

MainActivity.java:

```
package com.example.labcycle;

import android.annotation.SuppressLint;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;
import android.widget.EditText;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    EditText m1, m2, m3;
    Button btn;
    TextView t1, t2;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        m1 = findViewById(R.id.mark1);
        m2 = findViewById(R.id.mark2);
        m3 = findViewById(R.id.mark3);
        btn = findViewById(R.id.btncal);
        t1 = findViewById(R.id.textView4);
        t2 = findViewById(R.id.textView5);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                int n1 = Integer.parseInt(m1.getText().toString());
                int n2 = Integer.parseInt(m2.getText().toString());
```

```

        int n3 = Integer.parseInt(m3.getText().toString());
        int sums = n1 + n2 + n3;
        int avg = sums / 3;
        t1.setText("Sum is " + sums);
        t2.setText("Average is " + avg);
    }
});
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/mark1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="92dp"
        android:layout_marginBottom="43dp"
        android:ems="10"
        android:inputType="number"
        android:text="mark1"
        app:layout_constraintBottom_toTopOf="@+id/mark2"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText
        android:id="@+id/mark2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="91dp"
        android:layout_marginBottom="40dp"
        android:ems="10"
        android:inputType="number"
        android:text="mark2"
        app:layout_constraintBottom_toTopOf="@+id/mark3"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText

```

```

        android:id="@+id/mark3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="85dp"
        android:layout_marginBottom="48dp"
        android:ems="10"
        android:inputType="numberSigned"
        android:text="mark3"
        app:layout_constraintBottom_toTopOf="@+id/btnncal"
        app:layout_constraintEnd_toEndOf="parent" />

<Button
    android:id="@+id/btnncal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="182dp"
    android:layout_marginBottom="49dp"
    android:text="Button"
    app:layout_constraintBottom_toTopOf="@+id/textView4"
    app:layout_constraintEnd_toEndOf="parent" />

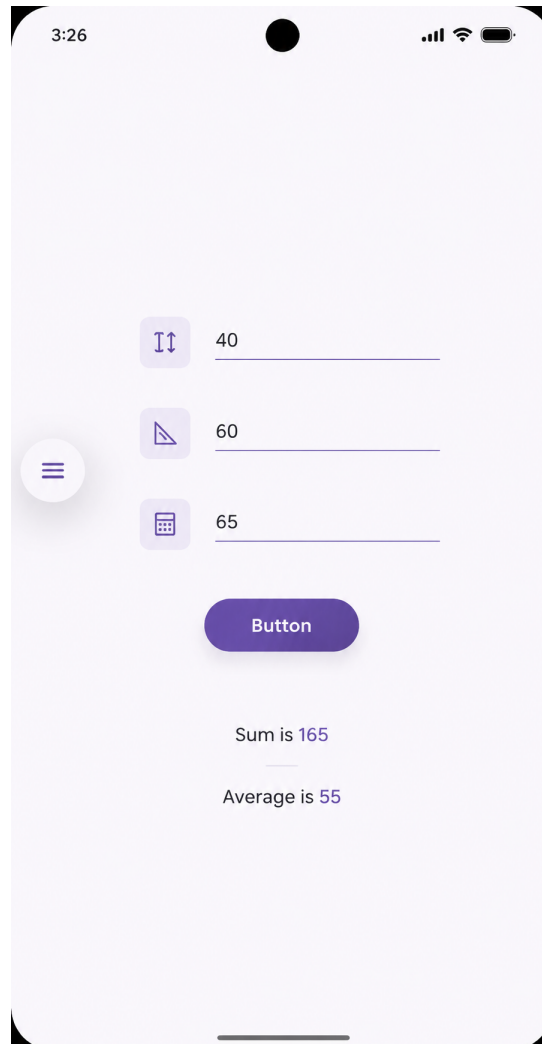
<TextView
    android:id="@+id/textView4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="202dp"
    android:layout_marginBottom="160dp"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="200dp"
    android:layout_marginBottom="120dp"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 3

Date: 29/06/2026

Program Title: Create suitable apps to design the different layouts of the android-studio.

MainActivity.java:

```
package com.example.labcycle;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    Button button1, button2, button3;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        button1 = findViewById(R.id.button1);
        button2 = findViewById(R.id.button2);
        button3 = findViewById(R.id.button3);

        button1.setOnClickListener(v ->
            Toast.makeText(this, "LinearLayout Button 1",
                Toast.LENGTH_SHORT).show());
        button2.setOnClickListener(v ->
            Toast.makeText(this, "LinearLayout Button 2",
                Toast.LENGTH_SHORT).show());
        button3.setOnClickListener(v ->
            Toast.makeText(this, "RelativeLayout Button",
                Toast.LENGTH_SHORT).show());
    }
}
```

activity__main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
```



```

        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

<!-- LinearLayout -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Linear Layout"
    android:textStyle="bold"
    android:textSize="18sp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="20dp">

    <Button
        android:id="@+id/button1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Button 1"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Button 2"/>
</LinearLayout>

<!-- RelativeLayout -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Relative Layout"
    android:textStyle="bold"
    android:textSize="18sp"/>

<RelativeLayout
    android:layout_width="match_parent"

```

```

        android:layout_height="120dp"
        android:background="#DDDDDD"
        android:layout_marginBottom="20dp">

        <TextView
            android:id="@+id/text1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Top Left"/>

        <Button
            android:id="@+id/button3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Bottom Right"
            android:layout_alignParentBottom="true"
            android:layout_alignParentEnd="true"/>
    </RelativeLayout>

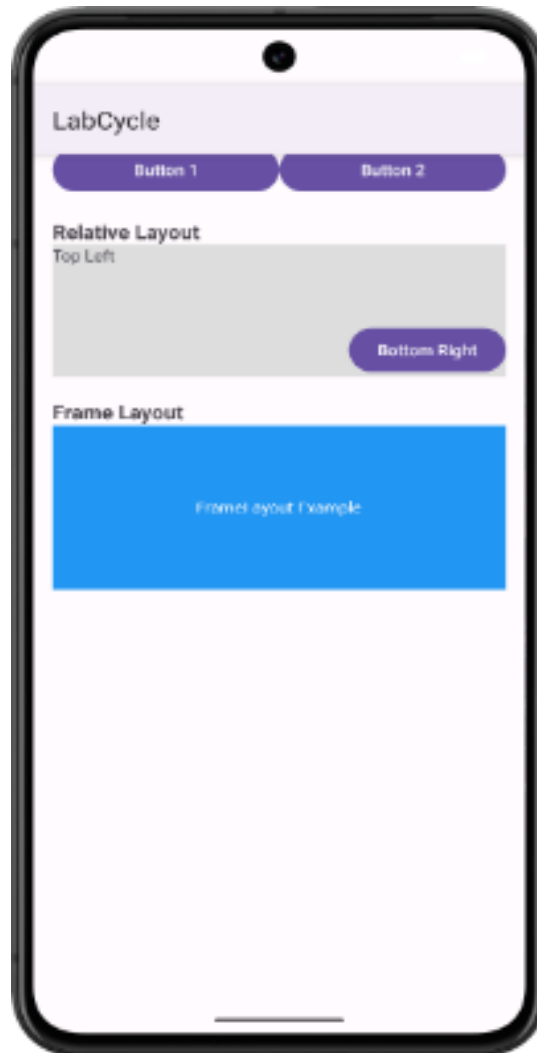
    <!-- FrameLayout -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Frame Layout"
        android:textStyle="bold"
        android:textSize="18sp"/>

    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:background="#2196F3">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="FrameLayout Example"
            android:textColor="#FFFFFF"
            android:layout_gravity="center"/>
    </FrameLayout>
</LinearLayout>
</ScrollView>

```

Output:



Program No: 4

Date: 29/06/2026

Program Title: Develop a simple calculator app.

MainActivity.java:

```
package com.example.pg04_calculator;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.TextView;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {
    TextView result;
    Button ad, sub, div, mult;
    EditText no1, no2;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        result = findViewById(R.id.result);
        ad = findViewById(R.id.ad);
        sub = findViewById(R.id.sub);
        div = findViewById(R.id.div);
        mult = findViewById(R.id.mult);
        no1 = findViewById(R.id.no1);
        no2 = findViewById(R.id.no2);

        ad.setOnClickListener(v -> {
            int a = Integer.parseInt(no1.getText().toString());
            int b = Integer.parseInt(no2.getText().toString());
            int res = a + b;
            result.setText(res+"");
        });

        sub.setOnClickListener(v -> {
            int a = Integer.parseInt(no1.getText().toString());
```

```

        int b = Integer.parseInt(no2.getText().toString());
        int res = a - b;
        result.setText(res+"");
    });

    mult.setOnClickListener(v -> {
        int a = Integer.parseInt(no1.getText().toString());
        int b = Integer.parseInt(no2.getText().toString());
        int res = a * b;
        result.setText(res+"");

    });

    div.setOnClickListener(v -> {
        int a = Integer.parseInt(no1.getText().toString());
        int b = Integer.parseInt(no2.getText().toString());
        if(b!=0) {
            int res = a / b;
            result.setText(res+"");
        }
        else
            result.setText("Error: Divsion / 0");
    });
}
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result: "
        android:textSize="18sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"

```

```

        app:layout_constraintHorizontal_bias="0.252"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

<TextView
    android:id="@+id/result"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.52"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<EditText
    android:id="@+id/no1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Enter First Number"
    android:inputType="text"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.437"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.228" />

<EditText
    android:id="@+id/no2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Enter Second Number"
    android:inputType="text"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.437"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.355" />

```

```

<Button
    android:id="@+id/ad"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Add"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.275"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.655" />

```

```

<Button
    android:id="@+id/sub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Subtract"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.64"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.655" />

```

```

<Button
    android:id="@+id/mult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Multiply"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.275"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.78" />

```

```

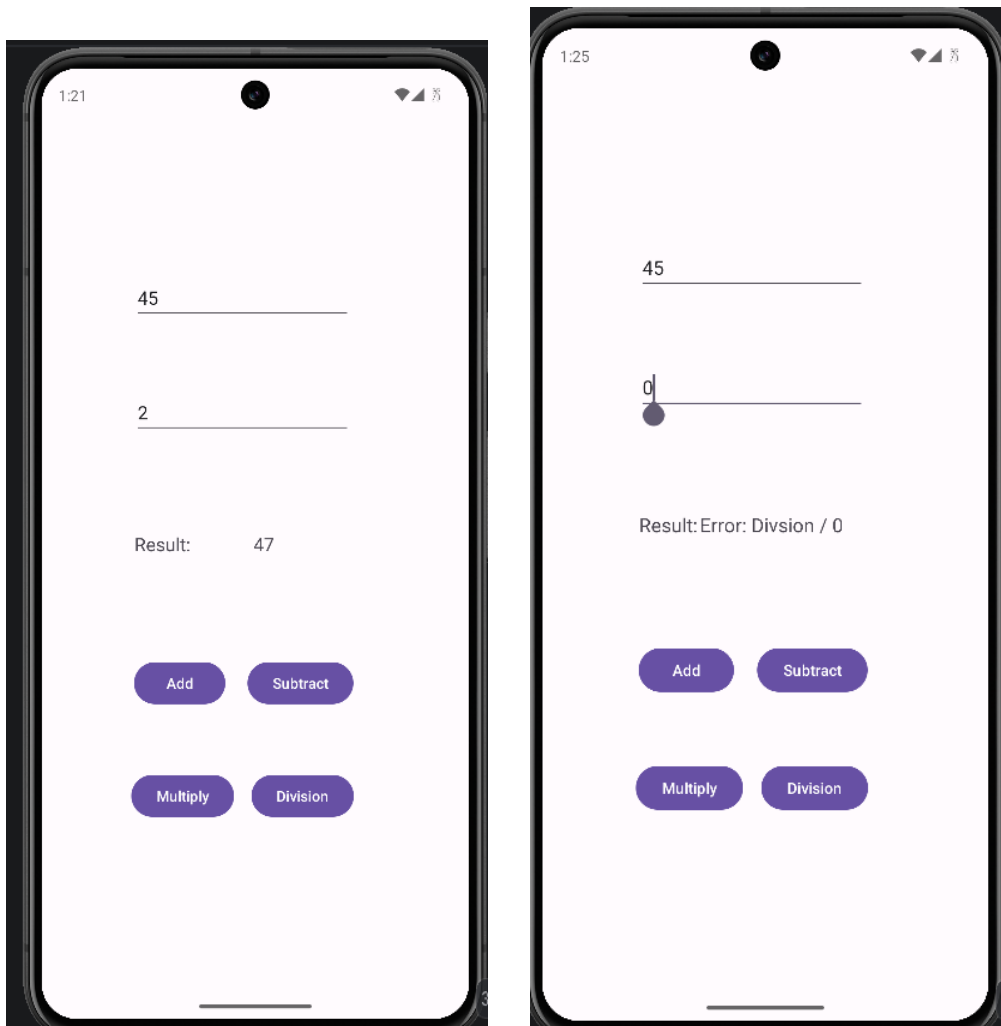
<Button
    android:id="@+id/div"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Division"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"

```

```
app:layout_constraintHorizontal_bias="0.646"  
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintTop_toTopOf="parent"  
app:layout_constraintVertical_bias="0.78" />
```

</androidx.constraintlayout.widget.ConstraintLayout>

Output:



Program No: 5

Date: 29/06/2026

Program Title: Create an app that changes text color when a button is clicked.

MainActivity.java:

```
package com.example.pg05_textcolor;

import android.graphics.Color;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.TextView;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    TextView name;
    Button color;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        color = findViewById(R.id.color);
        name = findViewById(R.id.name);
        color.setOnClickListener(v -> {
            name.setTextColor(Color.RED);
        });

    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```

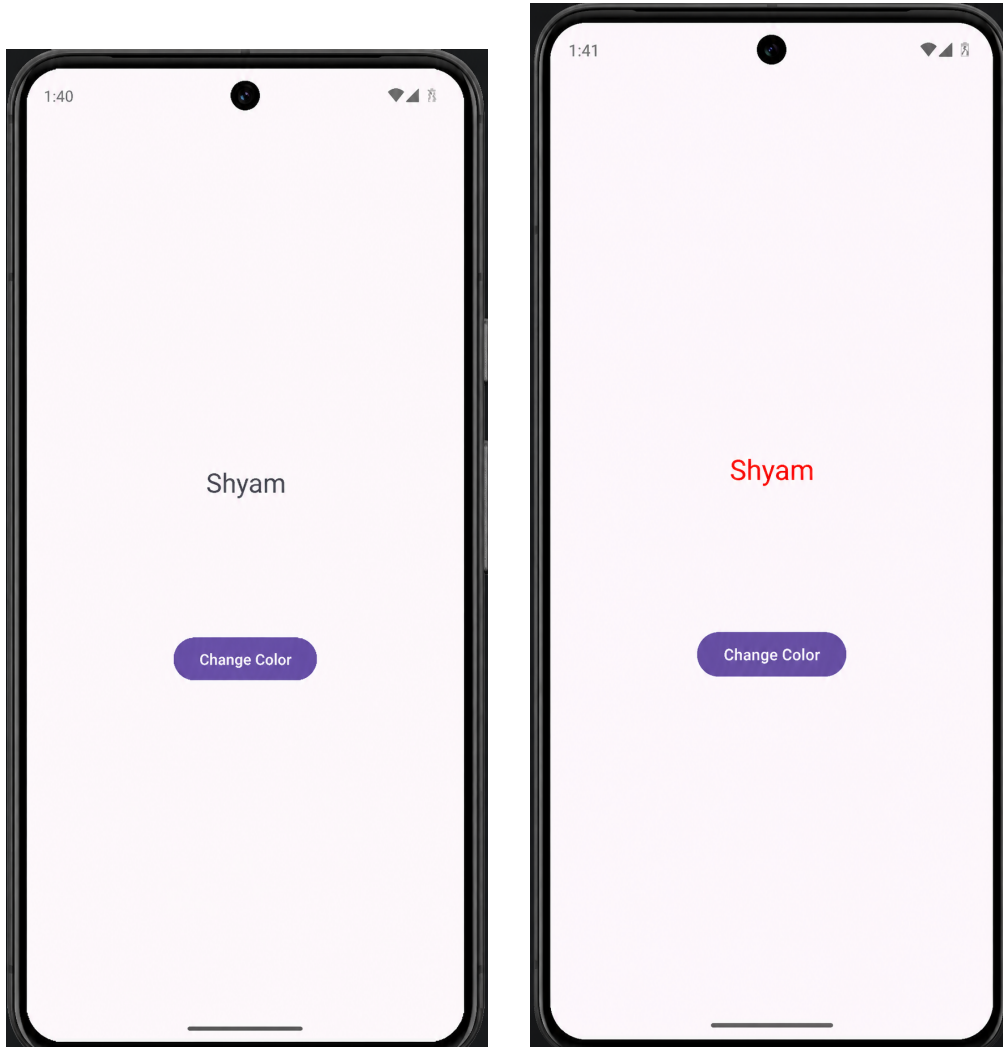
<TextView
    android:id="@+id/name"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Shyam"
    android:textSize="24sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.421" />

<Button
    android:id="@+id/color"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Change Color"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.613" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 6

Date: 29/06/2026

Program Title: Create an app that displays Date and Time in different formats using Linear Layout.

MainActivity.java:

```
package com.example.pg06_datetime;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Locale;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        final TextView textView1 = findViewById(R.id.textView1);
        final TextView textView2 = findViewById(R.id.textView2);
        final TextView textView3 = findViewById(R.id.textView3);
        final TextView textView4 = findViewById(R.id.textView4);
        Button button1 = findViewById(R.id.button1);
        button1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Date currentDateTime = new Date();
                SimpleDateFormat format1 = new SimpleDateFormat("EEEE, dd
MMM yyyy", Locale.getDefault());
                SimpleDateFormat format2 = new SimpleDateFormat("MM/dd/yyyy",
Locale.getDefault());
                SimpleDateFormat format3 = new SimpleDateFormat("HH:mm:ss
z", Locale.getDefault());
                SimpleDateFormat format4 = new SimpleDateFormat("hh:mm a",
Locale.getDefault());
                textView1.setText(format1.format(currentDateTime));
                textView2.setText(format2.format(currentDateTime));
                textView3.setText(format3.format(currentDateTime));
                textView4.setText(format4.format(currentDateTime));
            }
        })
    }
}
```

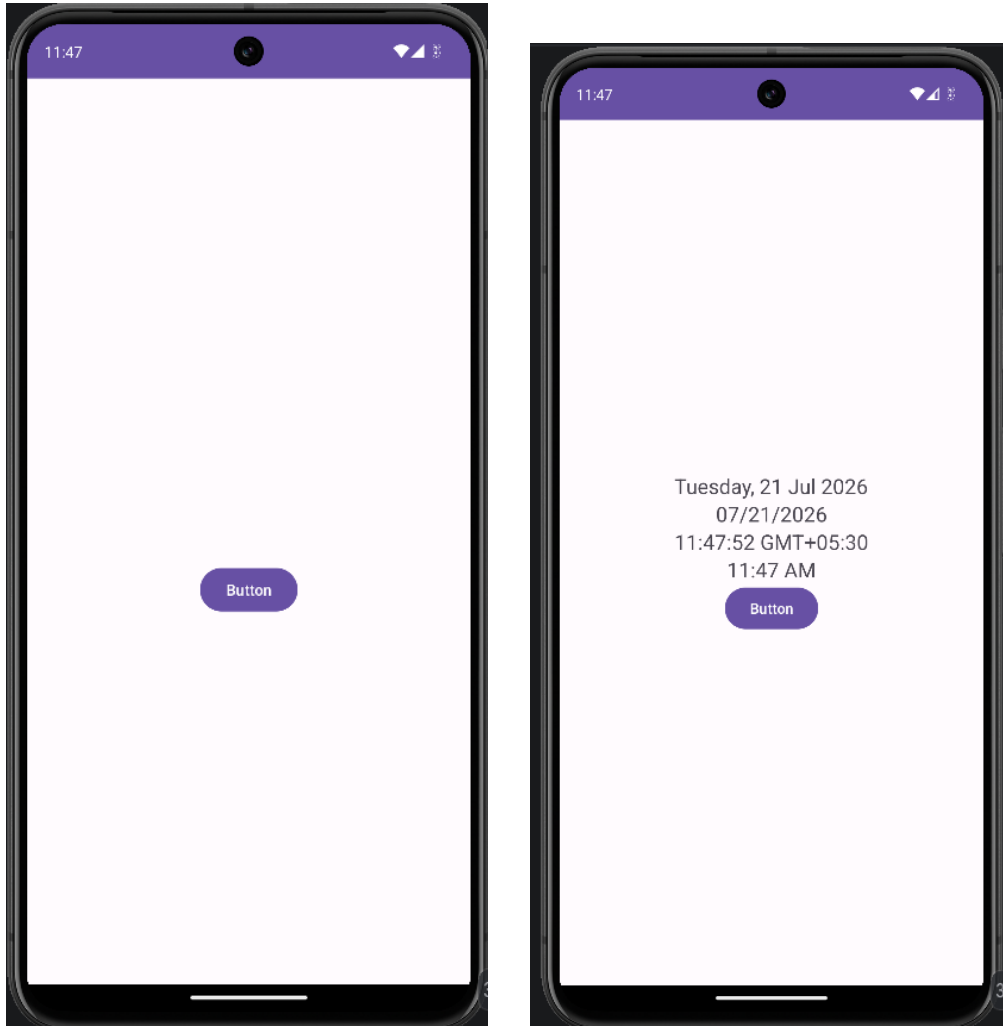
```
});  
}  
}
```

activity__main.xml:

```
<?xml version="1.0" encoding="utf-8"?>  
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"  
    xmlns:app="http://schemas.android.com/apk/res-auto"  
    xmlns:tools="http://schemas.android.com/tools"  
    android:id="@+id/main"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="vertical"  
    android:gravity="center"  
    tools:context=".MainActivity">  
  
    <TextView  
        android:id="@+id/textView1"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:textSize="20sp" />  
  
    <TextView  
        android:id="@+id/textView2"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:textSize="20sp" />  
  
    <TextView  
        android:id="@+id/textView3"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:textSize="20sp" />  
  
    <TextView  
        android:id="@+id/textView4"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:textSize="20sp" />  
    <Button  
        android:id="@+id/button1"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:text="Button" />
```

`</LinearLayout>`

Output:



Program No: 7

Date: 29/06/2026

Program Title: Create an app that displays a Toast on the launch of an activity.

MainActivity.java:

```
package com.example.pg07_launchtoast;

import android.os.Bundle;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        Toast.makeText(this, "Welcome Shyam", Toast.LENGTH_SHORT).show();
    }
}
```

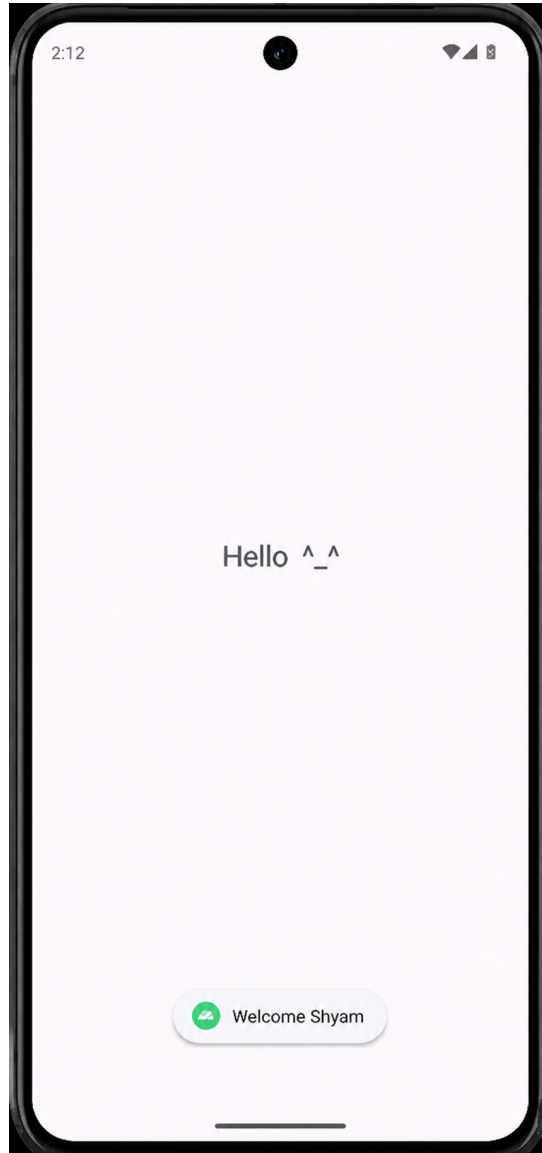
activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello ^_^"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 8

Date: 29/06/2026

Program Title: Create an app that creates an Alert Dialog Box.

MainActivity.java:

```
package com.example.pg08_alertbox;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AlertDialog;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.btn);
        btn.setOnClickListener(v -> {
            AlertDialog.Builder builder = new AlertDialog.Builder(MainActivity.this);
            builder.setTitle("Alert");
            builder.setMessage("What do you want to click?");

            builder.setPositiveButton("Yes", (dialog, which) -> {
                Toast.makeText(MainActivity.this, "You clicked Yes",
                    Toast.LENGTH_SHORT).show();
            });

            builder.setNegativeButton("No", (dialog, which) -> {
                Toast.makeText(MainActivity.this, "You clicked No",
                    Toast.LENGTH_SHORT).show();
            });

            builder.setNeutralButton("Cancel", (dialog, which) -> {
                dialog.dismiss();
            });

            builder.show();
        });
    }
}
```

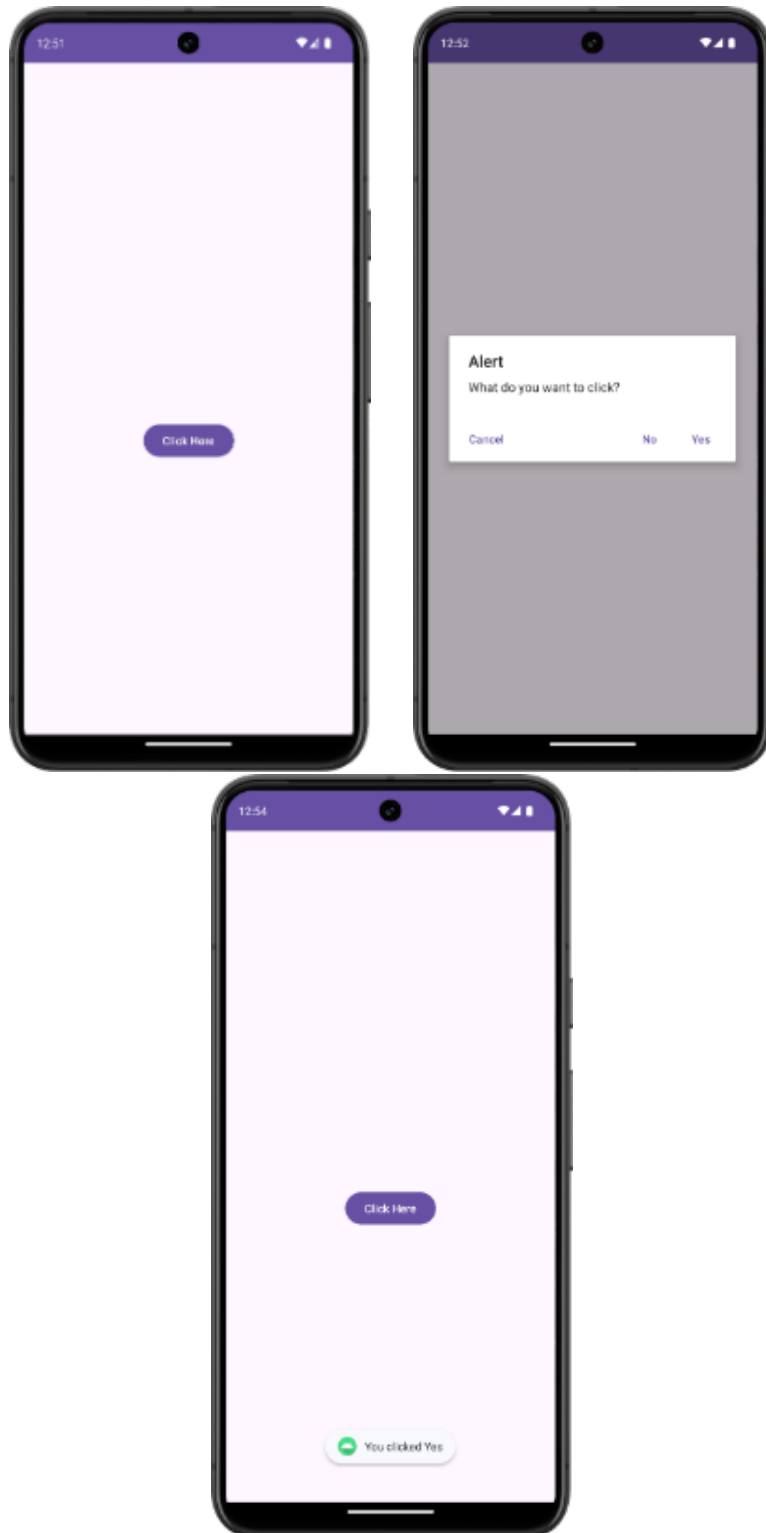
activity__main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Here"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.566" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 9

Date: 29/06/2026

Program Title: Create a Greeting card app that displays an image.

MainActivity.java:

```
package com.example.pg09_greetingcard;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="288dp"
        android:layout_height="393dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.495"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.328"
        app:srcCompat="@drawable/greetings" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```

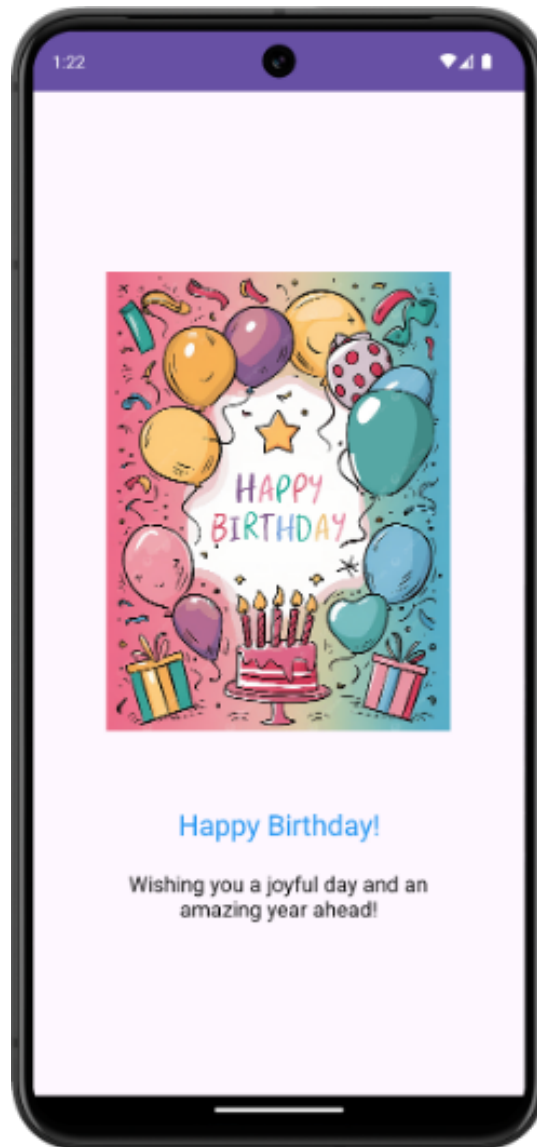
        android:text="Happy Birthday!"
        android:textColor="#2196F3"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.739" />

<TextView
    android:id="@+id/textView"
    android:layout_width="258dp"
    android:layout_height="59dp"
    android:text="Wishing you a joyful day and an amazing year ahead!"
    android:textAlignment="center"
    android:textColor="#020202"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.496"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.833" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program Title: Create an app that plays audio when we click on the button “Play Music” and plays video when we click on the button “Play Video”.

MainActivity.java:

```
package com.example.pg10_videomusic;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;
import android.widget.VideoView;
import android.net.Uri;
import android.media.MediaPlayer;
import android.widget.MediaController;

public class MainActivity extends AppCompatActivity {
    MediaPlayer mediaPlayer;
    VideoView videoView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnMusic = findViewById(R.id.music);
        Button btnVideo = findViewById(R.id.video);

        videoView = findViewById(R.id.videoView);
        videoView.setZOrderOnTop(true);

        mediaPlayer = MediaPlayer.create(this, R.raw.song);

        btnMusic.setOnClickListener(v -> {
            if (!mediaPlayer.isPlaying()) {
                mediaPlayer.start();
            }
        });

        btnVideo.setOnClickListener(v -> {
            Uri uri = Uri.parse("android.resource://" + getPackageName()
                + "/" + R.raw.video);
            videoView.setVideoURI(uri);
            MediaController controller =
```

```

        new MediaController(MainActivity.this);
        videoView.setMediaController(controller);
        controller.setAnchorView(videoView);
        videoView.start();
    });
}

@Override
protected void onDestroy() {
    super.onDestroy();
    if (mediaPlayer != null) {
        mediaPlayer.release();
        mediaPlayer = null;
    }
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <VideoView
        android:id="@+id/videoView"
        android:layout_width="348dp"
        android:layout_height="231dp"
        android:background="#000000"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.492"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.23" />

    <Button
        android:id="@+id/music"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

```



```

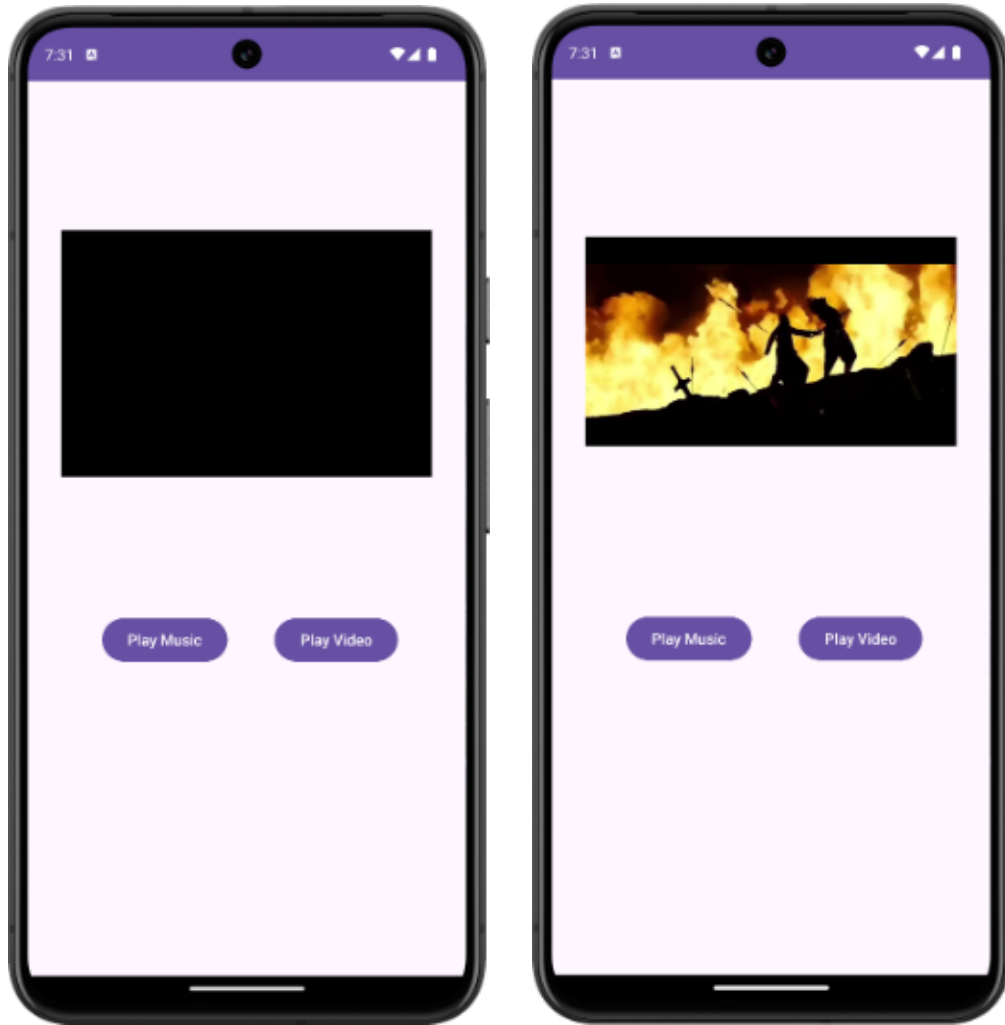
        android:text="Play Music"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.237"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.632" />

<Button
    android:id="@+id/video"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="8dp"
    android:text="Play Video"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.777"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.632" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 11

Date: 29/06/2026

Program Title: Create an app that plays video from the internet.

MainActivity.java:

```
package com.example.pg11_internetvideo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;
import android.widget.VideoView;
import android.net.Uri;
import android.widget.MediaController;

public class MainActivity extends AppCompatActivity {
    VideoView videoView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnVideo = findViewById(R.id.play);
        videoView = findViewById(R.id.videoView);
        videoView.setZOrderOnTop(true);

        btnVideo.setOnClickListener(v -> {
            Uri uri = Uri.parse("https://www.learningcontainer.com/"
                                + "wp-content/uploads/2020/05/"
                                + "sample-mp4-file.mp4");
            videoView.setVideoURI(uri);
            MediaController controller =
                new MediaController(MainActivity.this);
            videoView.setMediaController(controller);
            controller.setAnchorView(videoView);
            videoView.start();
        });
    }
}
```

activity__main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```

xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

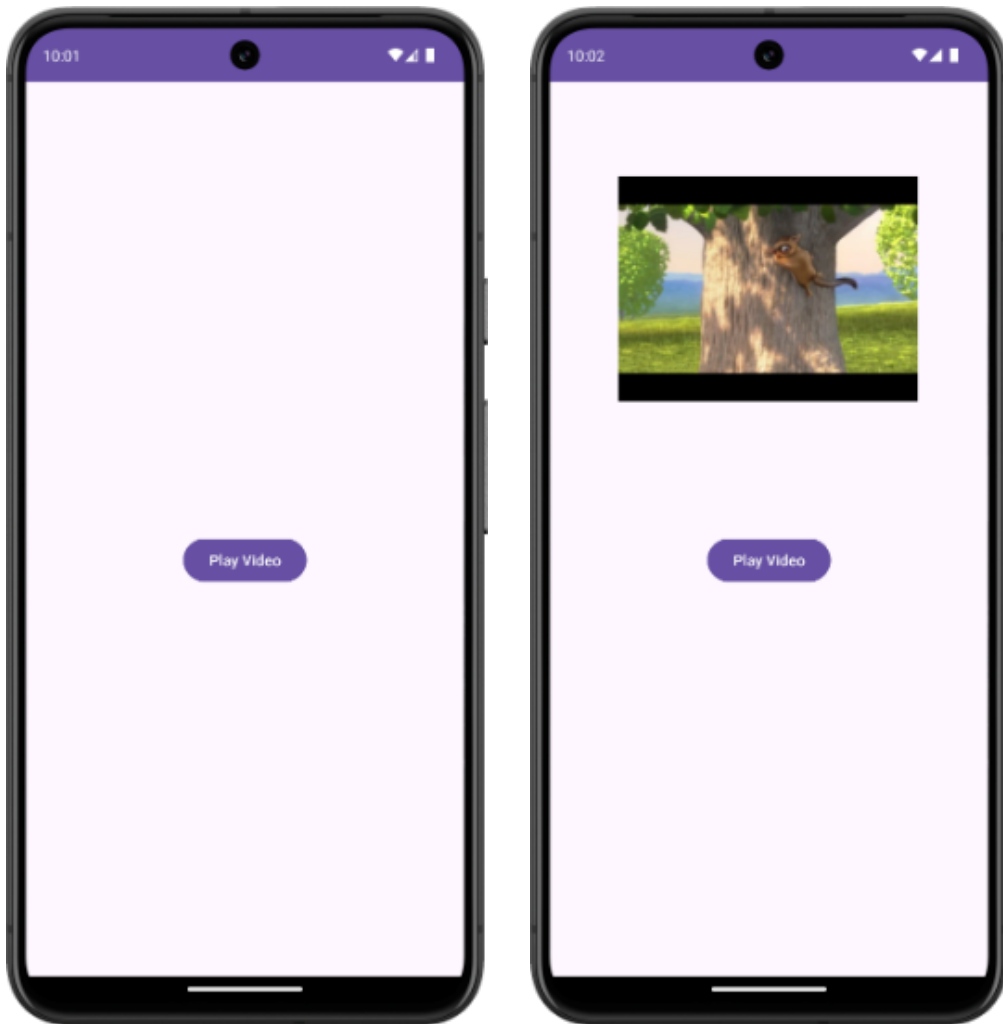
<VideoView
    android:id="@+id/videoView"
    android:layout_width="365dp"
    android:layout_height="211dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.493"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.141" />

<Button
    android:id="@+id/play"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Play Video"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.537" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 12

Date: 29/06/2026

Program Title: Create an app to demonstrate Options Menu.

MainActivity.java:

```
package com.example.pg12_optionsmenu;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.settings) {
            Toast.makeText(this, "Settings Selected",
                           Toast.LENGTH_SHORT).show();
        }
        else if (id == R.id.about) {
            Toast.makeText(this, "About Selected",
                           Toast.LENGTH_SHORT).show();
        }
        else if (id == R.id.exit) {
            finish();
        }
        return true;
    }
}
```

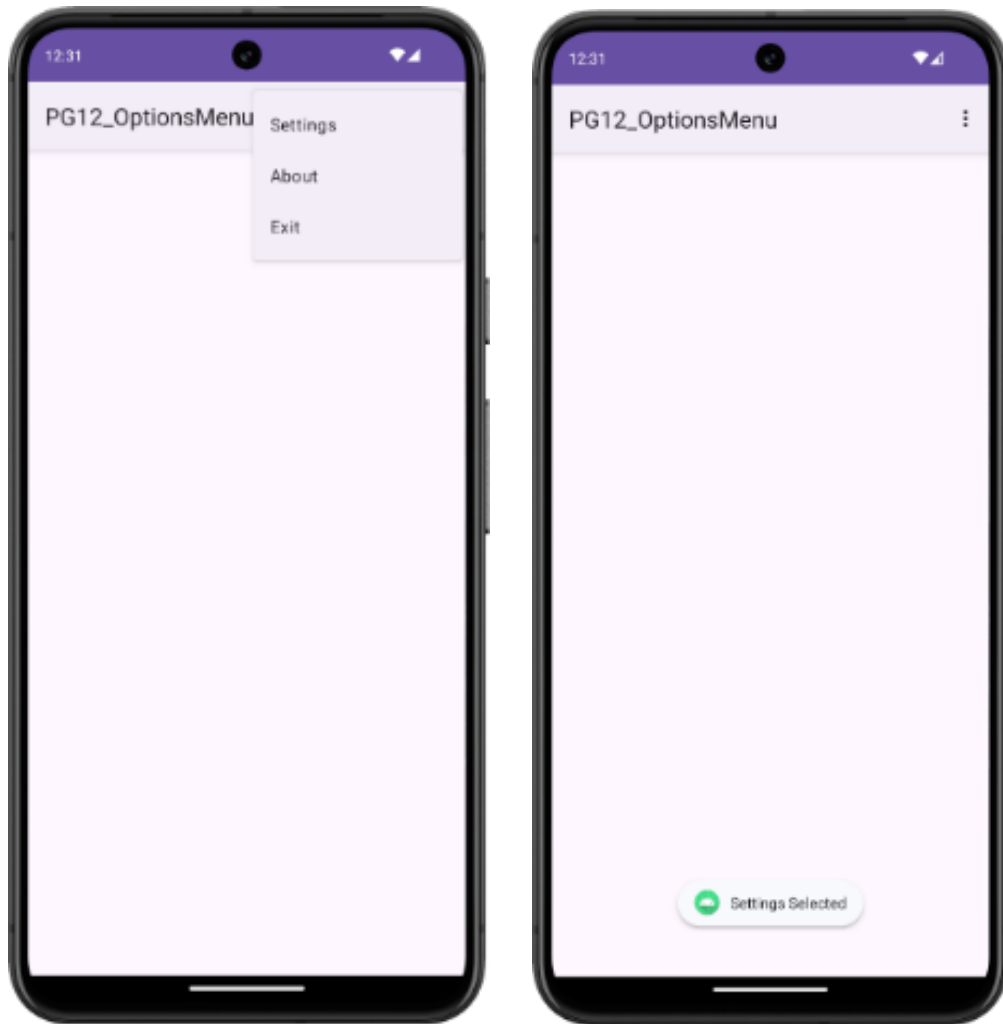
activity__main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"/>
```

menu.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/settings"
        android:title="Settings" />
    <item
        android:id="@+id/about"
        android:title="About" />
    <item
        android:id="@+id/exit"
        android:title="Exit" />
</menu>
```

Output:



Program No: 13

Date: 29/06/2026

Program Title: Create an app that changes the background color of an activity using Context Menu.

MainActivity.java:

```
package com.example.context_menu;

import android.graphics.Color;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.view.ContextMenu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.view.View;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    View layout;
    TextView tx1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        layout = findViewById(R.id.main);
        tx1 = findViewById(R.id.tx1);
        registerForContextMenu(tx1);
    }

    @Override
    public void onCreateContextMenu(ContextMenu menu, View v,
                                   ContextMenu.ContextMenuInfo menuinfo){
        super.onCreateContextMenu(menu, v, menuinfo);
        MenuInflater inflater = getMenuInflater();
        inflater.inflate(R.menu.menu_main, menu);
    }

    public boolean onContextItemSelected(MenuItem item) {
        if (item.getItemId() == R.id.red){
            layout.setBackgroundColor(Color.RED);
        }
        if (item.getItemId() == R.id.blue) {
            layout.setBackgroundColor(Color.BLUE);
        }
    }
}
```

```

    }
    if (item.getItemId() == R.id.green) {
        layout.setBackgroundColor(Color.GREEN);
    }
    if (item.getItemId() == R.id.yellow) {
        layout.setBackgroundColor(Color.YELLOW);
    }
    return false;
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/tx1"
        android:layout_width="90dp"
        android:layout_height="30dp"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.454"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.499" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

menu.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/red"
        android:title="RED" />
    <item
        android:id="@+id/blue"

```

```
        android:title="BLUE" />
    <item
        android:id="@+id/green"
        android:title="GREEN" />
    <item
        android:id="@+id/yellow"
        android:title="YELLOW" />
</menu>
```

Output:



Program No: 14

Date: 29/06/2026

Program Title: Create an app to demonstrate how radiobuttons and checkboxes can be used in android.

MainActivity.java:

```
package com.example.radiocheckboxbox;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;
import android.widget.RadioGroup;
import android.widget.RadioButton;
import android.widget.CheckBox;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    RadioGroup gender;
    CheckBox ten, twelve, UG, PG;
    Button submit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        gender = findViewById(R.id.gender);
        ten = findViewById(R.id.ten);
        twelve = findViewById(R.id.twelve);
        UG = findViewById(R.id.UG);
        PG = findViewById(R.id.PG);
        submit = findViewById(R.id.submit);

        submit.setOnClickListener(v ->{
            int selectedId = gender.getCheckedRadioButtonId();
            RadioButton rb = findViewById(selectedId);
            String gen = rb.getText().toString();
            StringBuilder education = new StringBuilder();

            if (ten.isChecked()) education.append("Tenth ");
            if (twelve.isChecked()) education.append("Plus Two ");
            if (UG.isChecked()) education.append("Undergraduate ");
            if (PG.isChecked()) education.append("Postgraduate ");
```

```

        Toast.makeText(this, gen + " | " + education,
                        Toast.LENGTH_LONG).show();
    });
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FB9E9E"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView4"
        android:layout_width="82dp"
        android:layout_height="23dp"
        android:text="Gender: "
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.124"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.114" />

    <RadioGroup
        android:id="@+id/gender"
        android:layout_width="247dp"
        android:layout_height="143dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.25"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.205">
        <RadioButton
            android:id="@+id/male"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"

```

```

        android:text="Male" />
<RadioButton
    android:id="@+id/female"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Female" />
<RadioButton
    android:id="@+id/tg"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="TG" />
</RadioGroup>

<CheckBox
    android:id="@+id/ten"
    android:layout_width="134dp"
    android:layout_height="43dp"
    android:text="X"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.155"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.527" />

<CheckBox
    android:id="@+id/UG"
    android:layout_width="133dp"
    android:layout_height="48dp"
    android:text="UG"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.154"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.71" />

<CheckBox
    android:id="@+id/twelve"
    android:layout_width="134dp"
    android:layout_height="48dp"
    android:text="XII"
    app:layout_constraintBottom_toBottomOf="parent"

```

```

        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.158"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.619" />

<Button
    android:id="@+id/submit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.468"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.915" />

<TextView
    android:id="@+id/textView"
    android:layout_width="90dp"
    android:layout_height="22dp"
    android:text="Qualification"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.13"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.452" />

<CheckBox
    android:id="@+id/PG"
    android:layout_width="133dp"
    android:layout_height="48dp"
    android:text="PG"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.154"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.808" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:

The image displays two side-by-side smartphone screens showing a form with gender and qualification options. The left screen shows the initial state with all options unselected. The right screen shows the state after selection, with 'Male', 'X', 'XII', and 'UG' selected, and a confirmation message at the bottom.

Left Screen (12:50):

Gender:

- ☐ Male
- ☐ Female
- ☐ TG

Qualification

- ☐ X
- ☐ XII
- ☐ UG
- ☐ PG

Submit

Right Screen (12:53):

Gender:

- ☒ Male
- ☐ Female
- ☐ TG

Qualification

- ☒ X
- ☒ XII
- ☒ UG
- ☐ PG

Submit

Male | Tenth Plus Two Undergraduate